
Midterm Exam 1

Last Name	First Name	Student ID # U
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Honor Code: This exam represents only my own work. I did not give or receive help from anyone.

Signature: _____

Instructions

- You may use the **instructor-provided reference sheet** (double-sided). You may add your own handwritten notes to this sheet. No other notes, textbooks, or personal reference materials.
- Calculator permitted (non-programmable, non-graphing).
- Write all answers in the space provided. If you need more room, use the extra page provided and indicate clearly.
- Show your work. Correct answers without supporting work may not receive full credit.
- Do not open this exam until instructed.

Part I — Multiple Choice**[20 pts, 2 pts each]**

Fill in the bubble next to the single best answer.

Q1. Which delay type is most likely to vary unpredictably with traffic load on a shared network?

- ☐ Transmission delay
- ☐ Propagation delay
- ☐ Processing delay
- ☐ Queuing delay

Q2. The minimum frame size requirement in CSMA/CD exists so that:

- ☐ Larger frames carry proportionally less header overhead
- ☐ A station is still transmitting when the collision signal returns, enabling detection
- ☐ Smaller frames increase the probability of collision
- ☐ The standard mandates it for backward compatibility only

Q3. The Shannon entropy of a source is best described as:

- ☐ The number of symbols in the source alphabet
- ☐ The probability of the most likely symbol
- ☐ The minimum average number of bits needed to represent one symbol losslessly
- ☐ The maximum data rate achievable on a noisy channel

Q4. In the TCP/IP protocol stack, the Link layer is best described as responsible for:

- ☐ Routing packets across multiple networks using IP addresses
- ☐ End-to-end reliable byte-stream delivery between processes
- ☐ Transferring frames between directly connected nodes on the same segment
- ☐ Converting bits into electrical, optical, or radio signals

Q5. When an Ethernet receiver computes a non-zero CRC remainder for a received frame, it:

- ☐ Sends a NAK to request retransmission
- ☐ Forwards the frame and logs the error in the switch table
- ☐ Attempts to correct the error using the remainder value
- ☐ Silently discards the frame

Q6. The bit error probability for a binary matched-filter receiver is $P_b = Q\left(\sqrt{2E_b/N_0}\right)$. As E_b/N_0 increases, P_b :

- ☐ Increases — higher energy means more interference
- ☐ Stays constant — BER depends only on bandwidth
- ☐ Decreases — the argument of Q grows, driving Q toward zero
- ☐ Increases exponentially — this is the “waterfall” effect

Q7. An Ethernet switch learns which host is on which port by:

- ☐ Sending periodic ARP requests to all hosts
- ☐ Receiving a registration packet when each host powers on
- ☐ Recording the source MAC address and arrival port of every incoming frame
- ☐ Reading MAC addresses from IP packet headers

Q8. WiFi (802.11) cannot use collision *detection* (as classic Ethernet does) primarily because:

- ☐ WiFi data rates are too high to detect collisions in time
- ☐ The 2.4 GHz and 5 GHz bands are too noisy
- ☐ A transmitting node cannot listen while transmitting; it cannot hear its own collision
- ☐ The minimum frame size requirement makes collision detection unnecessary

Q9. In CSMA/CD binary exponential backoff, after the *third* consecutive collision a station chooses a random wait uniformly from how many slot times?

- ☐ 3 (linear backoff)
- ☐ 4 (2^2)
- ☐ 8 (2^3)
- ☐ 16 (2^4)

Q10. Manchester encoding is used in 10 Mb/s Ethernet (rather than NRZ) primarily because:

- ☐ Manchester achieves twice the data rate of NRZ at the same bandwidth
- ☐ The guaranteed mid-bit transition enables clock recovery at the receiver
- ☐ NRZ signals cannot propagate over twisted-pair copper
- ☐ Manchester eliminates the need for a preamble in the Ethernet frame

Part II — Short Answer**[20 pts, 4 pts each]**

Show your work. Concise calculations and one-sentence explanations are appropriate.

SA1. [4 pts] A 500 KB file is sent over a single link with capacity $R = 10$ Mb/s. The link is 2000 km long ($v = 2 \times 10^8$ m/s).

(a) What is the transmission delay?

(b) What is the propagation delay?

SA2. [4 pts] A wireless channel has bandwidth $B = 1$ MHz and $\text{SNR} = 31$ (linear). A modem claims to achieve 4.5 Mb/s on this channel. Is this possible?

SA3. [4 pts] In slotted ALOHA with offered load $G = 0.5$ frames per slot, arrivals per slot are Poisson: $P(\text{exactly } k \text{ arrivals}) = \frac{G^k e^{-G}}{k!}$.

What fraction of slots (as a percentage) experience a collision? Use $e^{-0.5} \approx 0.6$.

SA4. [4 pts] Host A (IP 10.1.1.1, MAC AA:AA:AA:AA:AA:AA) wants to send a packet to Host B (IP 10.1.1.2). A's ARP cache is empty.

(a) What Ethernet destination MAC does A use for the ARP request? _____

(b) After B replies with MAC BB:BB:BB:BB:BB:BB, A sends the IP datagram.

Ethernet source MAC: _____ Ethernet dst MAC: _____

SA5. [4 pts] A transmitter outputs $P_t = 20$ dBm. The path has 3 dB of cable/connector loss and 100 dB of free-space path loss. Receiver sensitivity is $P_{min} = -85$ dBm.

(a) What is P_r in dBm?

(b) What is the link margin? Is the link viable?

Part 3: Store-and-Forward in the Cubicle Maze [20 pts]

Consider a 5×5 grid of office cubicles, labeled (r, c) with $r, c \in \{0, 1, 2, 3, 4\}$. Occupants communicate by knocking on shared walls. The transmission rate is $R = 2$ bits/second and the propagation delay through one wall is $\tau = 0.2$ s. Each cubicle uses *store-and-forward*: the occupant waits for the complete message before relaying it. Messages travel along the shortest path.

(a) [10 pts] **Message format design.** Design a complete message format for this system: list every header field required, justify why each field is necessary, specify the minimum number of bits for each field, and compute the total message length L assuming an 8-bit payload.

(b) [10 pts] **End-to-end delay.** Person at $(0, 0)$ sends a message to person at $(0, 4)$ along the path $(0, 0) \rightarrow (0, 1) \rightarrow (0, 2) \rightarrow (0, 3) \rightarrow (0, 4)$. Using your message length L from part (a):

- (i) How many hops does the message take?
- (ii) What is the transmission delay per hop? The propagation delay per hop?
- (iii) What is the total end-to-end delay? (No queuing; no other traffic.)
- (iv) At what time does the *first bit* of the message arrive at $(0, 4)$? How does this compare to the total delay, and what does the difference represent?

Part 4: Error Control Coding & CRC [20 pts]

A CRC code uses generator polynomial $G(x) = x^3 + x + 1$ (bit string: 1011). Message words are $k = 3$ bits; $r = 3$ check bits are appended, giving $n = 6$ -bit codewords.

(a) [6 pts] Find the missing codeword for message 101. Show your work.

Seven of the eight valid codewords are listed below; one is missing.

Message	Codeword
000	000000
001	001011
010	010110
011	011101
100	100111
101	
110	110001
111	111010

(b) [8 pts] Determine d_{min} for this code. What are its error detection and error correction capabilities?

(c) [6 pts] The received word 001111 arrives. The receiver performs minimum-distance decoding (single-error correction). Which codeword does it choose, and what message is delivered? Does the receiver make an error here? Under what circumstances would a receiver like this make a mistake with this code?

Part 5: Modulation, Entropy, and Noise [20 pts]

Consider a **5-QAM** constellation: center point $s_0 = (0, 0)$ with probability $1/2$, and four corner points $(\pm A, \pm A)$ each with probability $1/8$. For comparison, a standard **4-QAM** constellation has four points at $(\pm A, \pm A)$, each with probability $1/4$.

Useful: $P_{avg} = \sum_i p_i(x_i^2 + y_i^2)$; $d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.

(a) [8 pts] Compute the entropy $H(X)$ for both the 5-QAM and 4-QAM sources. Show all terms. How do they compare?

(b) [6 pts] Determine d_{min} for each constellation in terms of A . Identify the pair of points that achieves the minimum distance in each case. At the same amplitude A , which constellation has greater noise immunity?

(c) [6 pts] Compute P_{avg} for each constellation in terms of A^2 . If both constellations are scaled to the *same* average power P , which has the larger d_{min} ? Is 5-QAM a better design than 4-QAM?

Extra Work Here

Clearly indicate which problem this work belongs to.